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Industrial Internship Report on :

Turbofan Engine Remaining Useful Life (RUL) Prediction

Prepared by :

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Executive Summary

This report presents the work completed during the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt Ltd (UCT).

The project focuses on predicting the **Remaining Useful Life (RUL)** of turbofan engines using machine learning and deep learning techniques. The goal is to enable predictive maintenance and reduce unexpected failures in industrial systems.

During this internship, I developed multiple models including Random Forest, Neural Networks, and Convolutional Neural Networks (CNN). Among these, the CNN model performed best due to its ability to capture temporal patterns in sensor data.

This internship provided valuable exposure to real-world industrial problems and helped me gain practical experience in data preprocessing, feature engineering, and model building.

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1 Preface

This internship spanned 6 weeks and focused on solving a real-world predictive maintenance problem.

Predictive maintenance is critical in industries where equipment failure can lead to high costs and safety risks. This project provided hands-on experience in applying machine learning to such problems.

The internship was well-structured with weekly goals including:

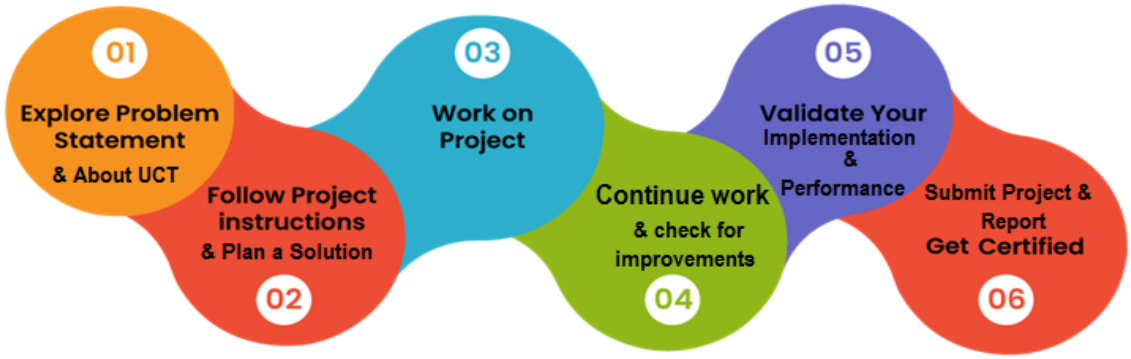
- Data understanding
- Model development
- Performance evaluation
- Final report preparation

I learned how to work with real-world datasets, handle large data, and build efficient ML models.

I would like to thank:

- Upskill Campus
- UniConverge Technologies Pvt Ltd
- The IoT Academy

for providing this opportunity.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

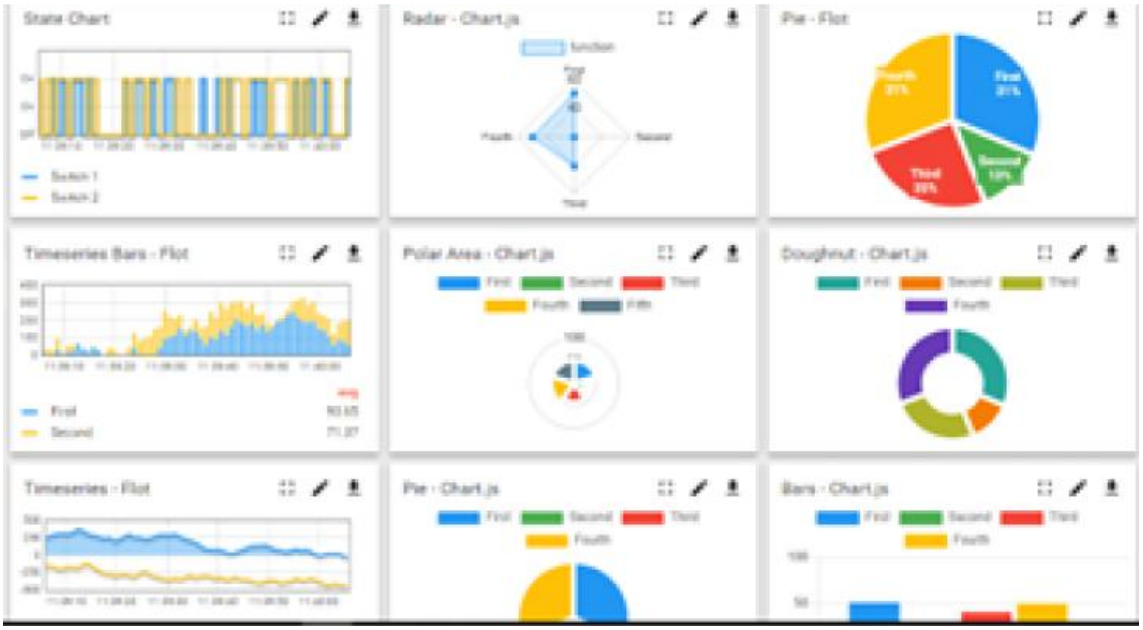
i. UCT IoT Platform (UCT Insight)

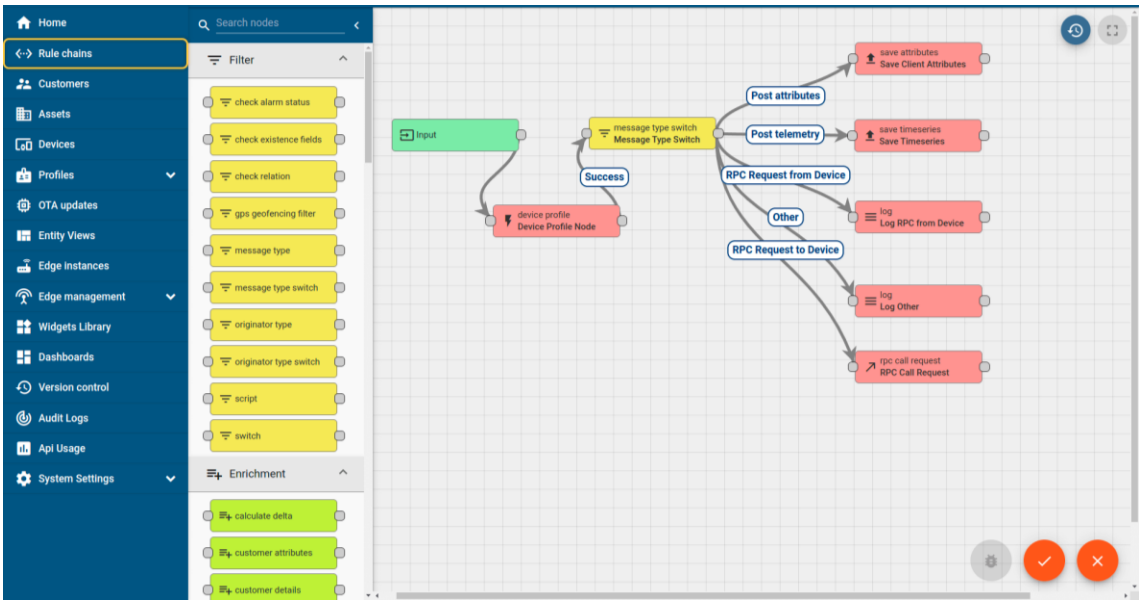
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

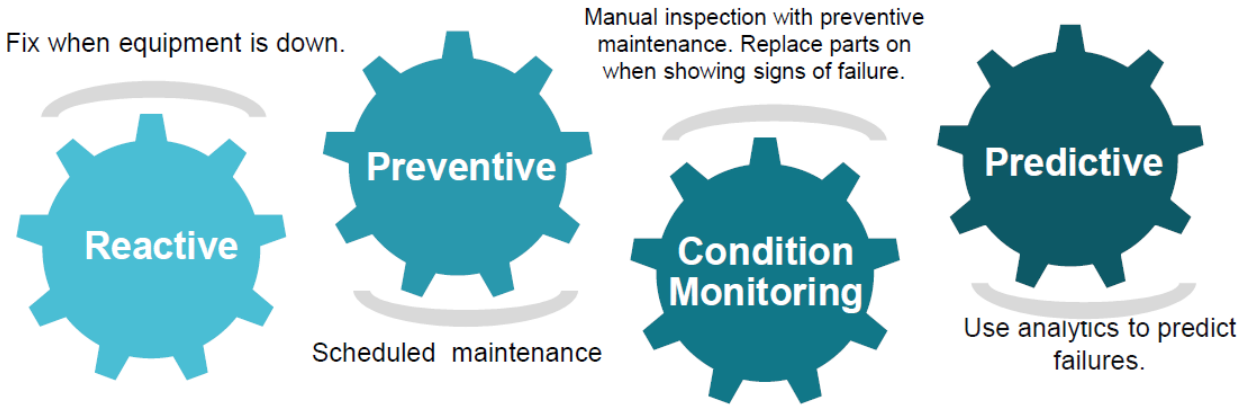
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

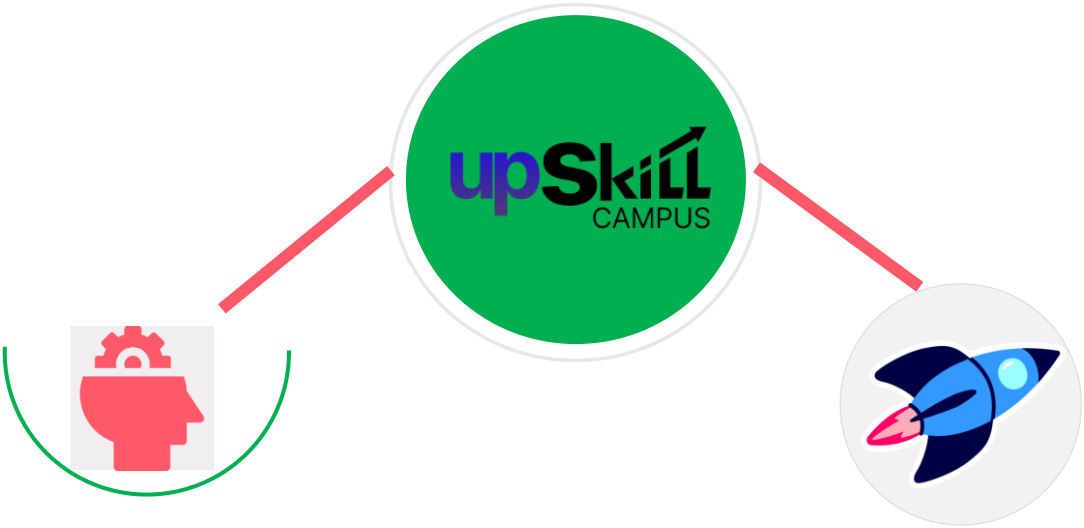
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

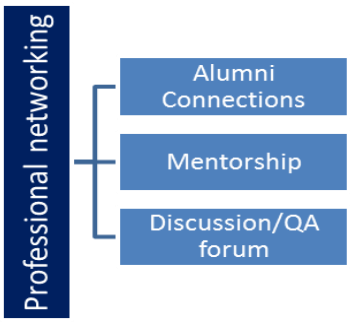
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

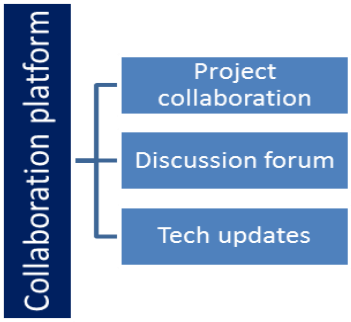


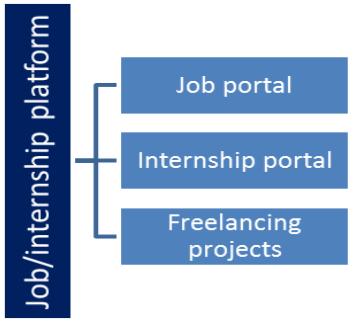
Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/







2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The main objective of this project is to develop a predictive model that can estimate the Remaining Useful Life (RUL) of turbofan engines. This helps industries perform maintenance before failure occurs, thereby improving efficiency and reducing downtime.

2.5 Reference

[1] NASA Ames Research Center, "Turbofan Engine Degradation Simulation Dataset (C-MAPSS)", Available: https://data.nasa.gov/

[2] Scikit-learn Documentation, "Machine Learning in Python", Available: https://scikit-learn.org/

[3] TensorFlow Documentation, "Deep Learning Framework", Available: https://www.tensorflow.org/

2.6 Glossary

Terms	Acronym
RUL	Remaining Useful Life
PdM	Predictive Maintenance
IoT	Internet of Things
CNN	Convolutional Neural Network
RMSE	Root Mean Squared Error
ANN	Artificial Neural Network

3 Problem Statement

The problem addressed in this project is to predict the number of operational cycles remaining before a turbofan engine fails. The dataset consists of multiple sensor readings recorded over time for different engines.

Since the data is sequential and time-dependent, the challenge lies in capturing patterns that indicate degradation. Accurate prediction of RUL allows industries to shift from reactive maintenance to predictive maintenance, leading to better resource utilization and cost savings.

4 Existing and Proposed solution

Traditional maintenance strategies are generally based on fixed schedules or reactive approaches, where maintenance is performed after a failure occurs. These methods are inefficient and often lead to unnecessary costs or unexpected breakdowns.

- **Existing Solutions**
- Traditional maintenance (time-based)
- Reactive maintenance (after failure)
- Limited ML-based approaches

Limitations:

- High cost
- Low accuracy
- No real-time insights

- **Proposed Solution**

  The proposed solution uses machine learning techniques to analyze historical sensor data and predict engine failure in advance. By training models on past data, the system learns patterns of degradation and can estimate the remaining life of the engine more accurately.

  This approach improves decision-making and provides a data-driven method for maintenance planning.

Develop ML/DL models to predict RUL using:

- Sensor data
- Historical engine degradation patterns

Approach:

- Data preprocessing
- Feature engineering
- Model training
- Performance evaluation

4.1 Code Submission:

https://github.com/OmkarDeshmukh001/upskillcampus/blob/main/RUL_CNN_Final.ipynb

4.2 Report Submission:

https://github.com/OmkarDeshmukh001/upskillcampus/blob/main/Project6_Pdm_TurbofanEngine_Omkar_USC_UCT.pdf

5 Proposed Design/ Model

The system is designed as a data-driven pipeline where raw sensor data is first preprocessed to remove noise and inconsistencies. Feature engineering techniques are applied to extract meaningful information that can improve model performance.

The processed data is then used to train different machine learning models such as Random Forest and Neural Networks. Additionally, a Convolutional Neural Network (CNN) is implemented to capture temporal dependencies in the data.

Among all models, the CNN model provides superior performance due to its ability to learn patterns across time sequences. The final output of the system is the predicted Remaining Useful Life of each engine.

5.1 Models Used

- Random Forest
- Neural Network
- CNN

5.2 High Level Diagram

Raw Sensor Data (NASA Dataset)

↓

Data Preprocessing

(Cleaning, Normalization)

↓

Feature Engineering

(RUL Calculation)

↓

Model Training

(RF / NN / CNN)

↓

Model Evaluation

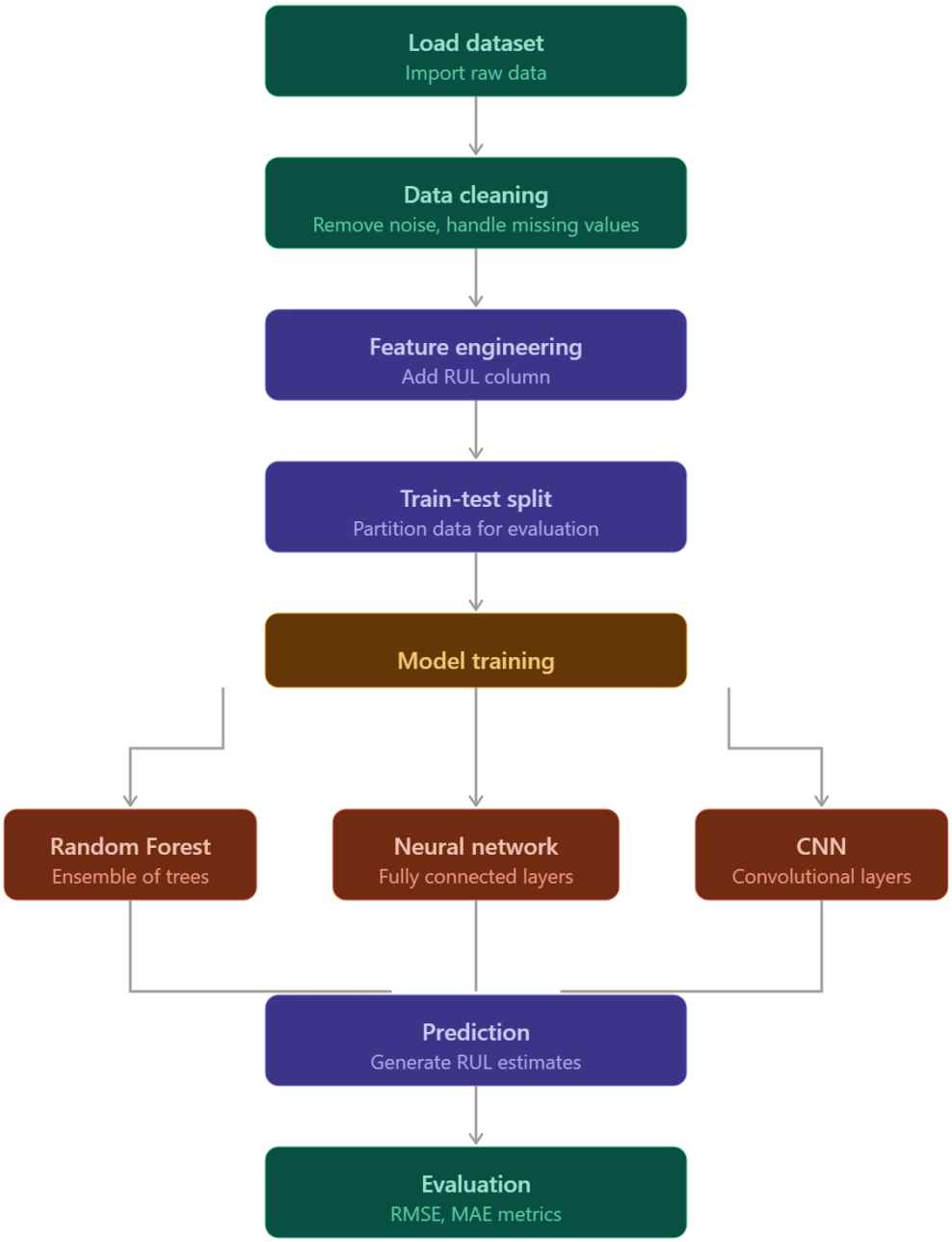
(RMSE, MAE)

↓

RUL Prediction Output

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.3 Low Level Diagram



5.4 Interfaces

The system interfaces define how different components interact and exchange data within the predictive maintenance pipeline. The primary interface is between the dataset and the preprocessing module, where raw sensor readings are converted into structured and usable formats.

Another key interface exists between the preprocessing stage and the machine learning models. Here, the processed feature vectors are passed as input to the models for training and prediction. The models then output predicted RUL values, which are further evaluated using performance metrics.

The system follows a data flow architecture where information moves sequentially through each stage without feedback loops. Since the project is implemented using Python and Jupyter Notebook, the interaction between modules is handled through function calls and data structures such as arrays and dataframes.

Although no external communication protocols are used in this project, the system can be extended to support APIs for real-time prediction. The current implementation is modular, allowing easy integration with web frameworks or industrial monitoring systems.

6 Performance Test

The performance of the models is evaluated using standard regression metrics such as Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). These metrics help in understanding how close the predicted values are to the actual values.

The CNN model achieved better accuracy compared to Random Forest and traditional Neural Networks. This is because CNNs are effective in handling sequential data and identifying complex patterns.

Some constraints considered during testing include dataset size, computational complexity, and model training time. Despite these challenges, the models were able to deliver reliable predictions.

6.1 Test Plan / Test Cases

The test plan for this project was designed to evaluate the performance and reliability of different machine learning models in predicting the Remaining Useful Life (RUL) of turbofan engines. The dataset was divided into training and testing sets to ensure unbiased evaluation. Multiple models, including Random Forest, Artificial Neural Networks, and Convolutional Neural Networks, were trained on the same dataset to enable a fair comparison.

Test cases were defined based on the model's ability to predict RUL accurately across different engine cycles. The evaluation focused on how well the models generalized to unseen data and how consistently they performed across varying operating conditions. Special attention was given to edge cases where the engine was close to failure, as accurate prediction in such scenarios is critical for real-world applications.

6.2 Test Procedure

The testing procedure involved a systematic approach starting with data preprocessing, where missing values were handled and features were normalized. The dataset was then split into training and testing subsets. Each model was trained using the training data and evaluated on the testing data.

After training, predictions were generated for the test dataset. These predictions were compared with the actual RUL values using standard evaluation metrics such as Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). The models were also analyzed in terms of training time and computational efficiency. This procedure ensured that the evaluation was consistent and comparable across all models.

6.3 Performance Outcome

The performance evaluation indicated that the Convolutional Neural Network (CNN) model achieved better accuracy compared to Random Forest and traditional Neural Network models. The CNN model was able to capture temporal patterns in the sensor data more effectively, resulting in improved prediction accuracy.

The Random Forest model provided stable results but lacked the ability to model sequential dependencies in the data. The Neural Network model showed moderate performance but required careful tuning of parameters. Overall, the CNN model demonstrated the best balance between accuracy and generalization.

These results highlight the importance of using deep learning techniques for time-series prediction problems, especially in predictive maintenance applications where accurate forecasting is essential.

7 My learnings

This internship provided significant learning in both technical and practical aspects. I gained hands-on experience in data preprocessing, machine learning model development, and performance evaluation.

I also learned how to approach real-world problems systematically and implement solutions using appropriate tools and techniques. The experience has strengthened my understanding of predictive maintenance and its applications in industry.

8 Future work scope

The project can be further enhanced by implementing advanced time-series models such as Long Short-Term Memory (LSTM) networks, which are specifically designed for sequential data.

Additionally, the model can be deployed using web frameworks like Flask or FastAPI to create a real-time prediction system. A user-friendly dashboard can also be developed using Streamlit to visualize predictions and insights.